

Report of Project 6: Equilibrium and Flamelet Modelling of Non-Premixed Turbulent Flames

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1 Equilibrium Model

1.1 PDF Table

1.1.1 Against Mixture Fraction – QA(3a)

Temperature, density, and mass fractions are dependent on the mixture fraction and the variance.
Outputs vs Mixture Fraction at Set Variance

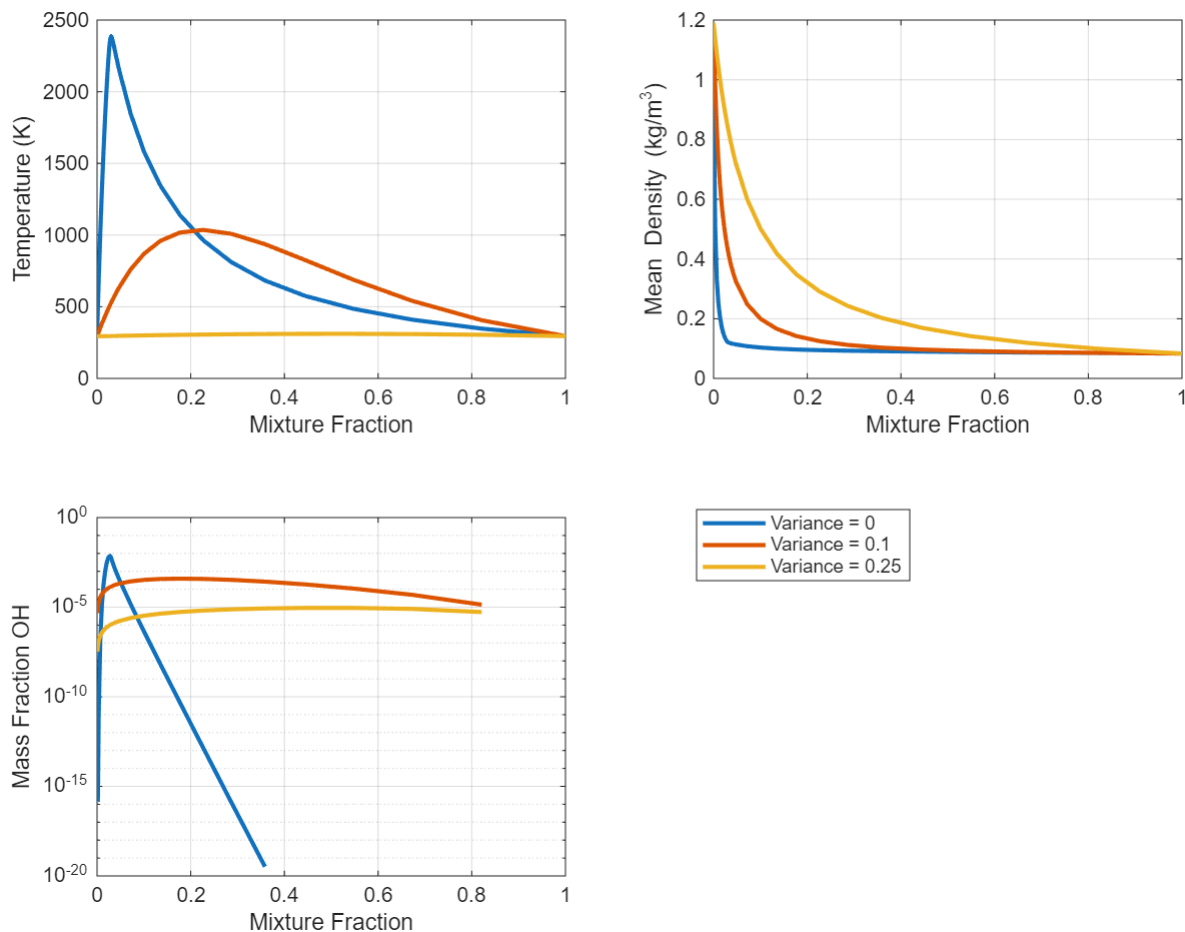


Figure 1 Temperature, Mean Density, and Mass fraction of OH Over Mixture Fraction

Fig. 1 shows the temperature, mean density, and mass fraction of OH as a function of mixture fraction at set variances of 0, 0.1, and 0.25. Temperature changes less over the mixture fraction as the variance increases as there is a large temperature spike with 0 variance, and there is essentially no change in temperature with a variance of 0.25. It can be shown in Fig. 1 that the density is larger for larger variances at set mixture fractions, but they all follows the same exponential decay trend over mixture fraction. The peak mass fraction of OH is larger for smaller variances; however, at 0 variance, it follows a slightly different trend of OH being used up quickly as mixture fraction increases compared to with a variance of 0.1 and 0.25 where the mass fractions are most constant.

1.1.2 Against Scaled Variance – QA(3b)

The stoichiometric mixture fraction of the hydrogen-air flame is 0.112. This was calculated from the mole fractions of H_2 , O_2 , N_2 , and H_2O being 1, 0.20759, 0.78091, and 0.0115 respectively. Hydrogen is the fuel and the other molecules are in the oxidizer (air).

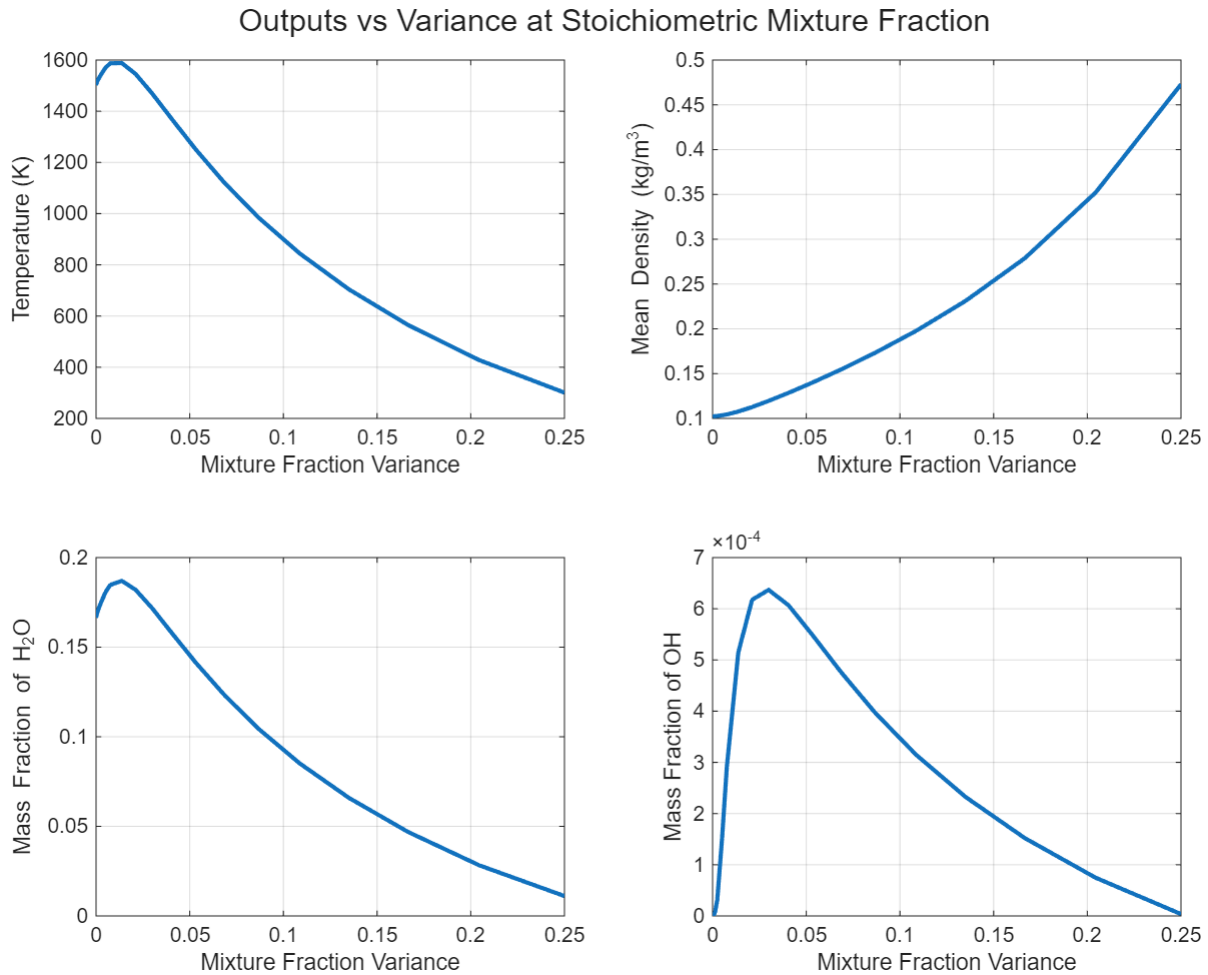


Figure 2 Temperature, Mean Density, Mass Fraction of OH, and Mass fraction of H_2O Over Mixture Fraction Variance

Fig. 2 shows the temperature, mean density, and mass fraction of both H_2O and OH over the mixture fraction variance for the stoichiometric mixture fraction of 0.112. Temperature, mass fraction of H_2O , and mass fraction of OH all follow the same trend of peaking at a mixture fraction variance of between 0 and 0.05 then following an exponential decay trend. Mean density has the opposite trend of an exponential growth as mixture fraction variance increases.

1.2 Variance Importance - QA(5)

The model constant C_g can be set to zero in Fluent which sets the production of scalar variance to zero. This is done to help see the influence of mixture fraction fluctuations by comparing the non-zero variance (base) model to the zero-variance model as well as with the experimental results.

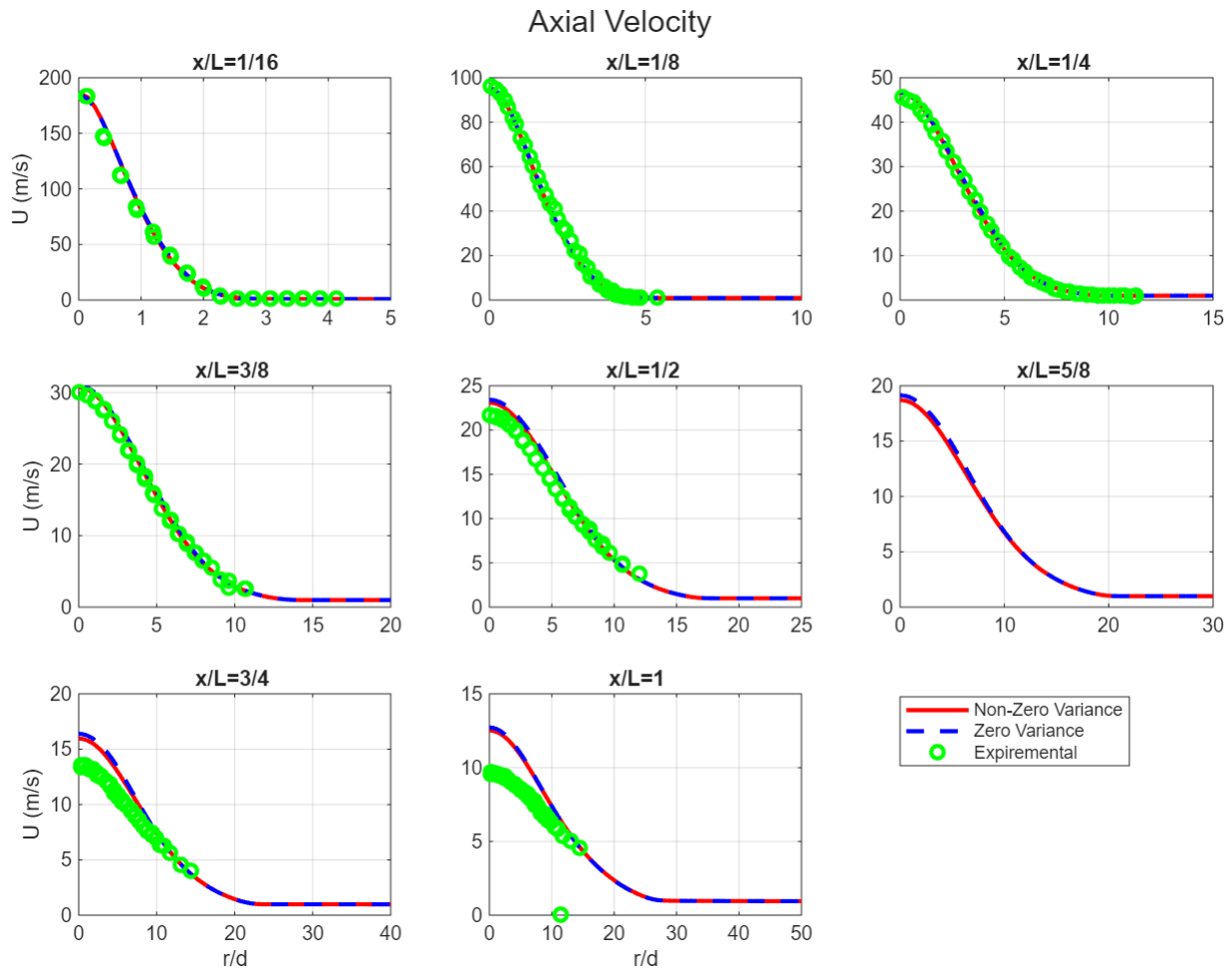


Figure 3 Axial Velocity versus r/d at Differing x/L

Fig. 3 shows the axial velocity as a function of radial distance at differing horizontal distance locations. The non-zero variance model and the zero variance model both follow very similar paths with the zero variance model having slightly larger axial velocity as the horizontal distance increases. The experimental results follow both models well; however, the correlation gets less at

an increasing horizontal distance as the models overestimate the axial velocity, especially near the centerline of the flame.

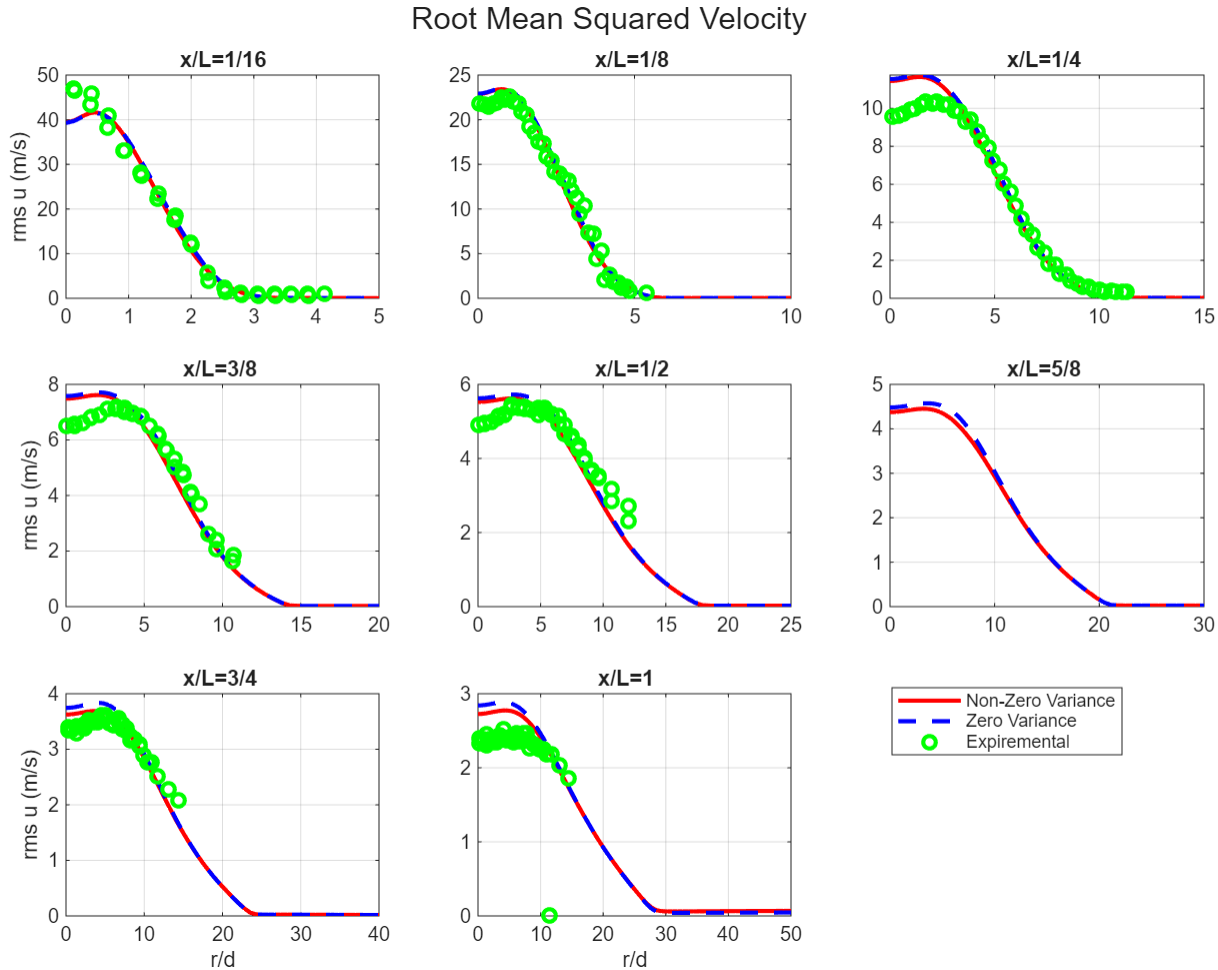


Fig. 4 shows the root mean squared velocity as a function of radial distance at differing horizontal distance locations. The non-zero variance model and the zero variance model both follow very similar paths with the zero variance model having slightly larger RMS velocity as the horizontal distance increases especially near the centerline of the flame. The experimental results are consistently under the model results which means the models are overestimating the results, especially near the centerline. The exception to this is near the flame inlet where at the centerline (the flame inlet), the models underestimate the RMS velocity.

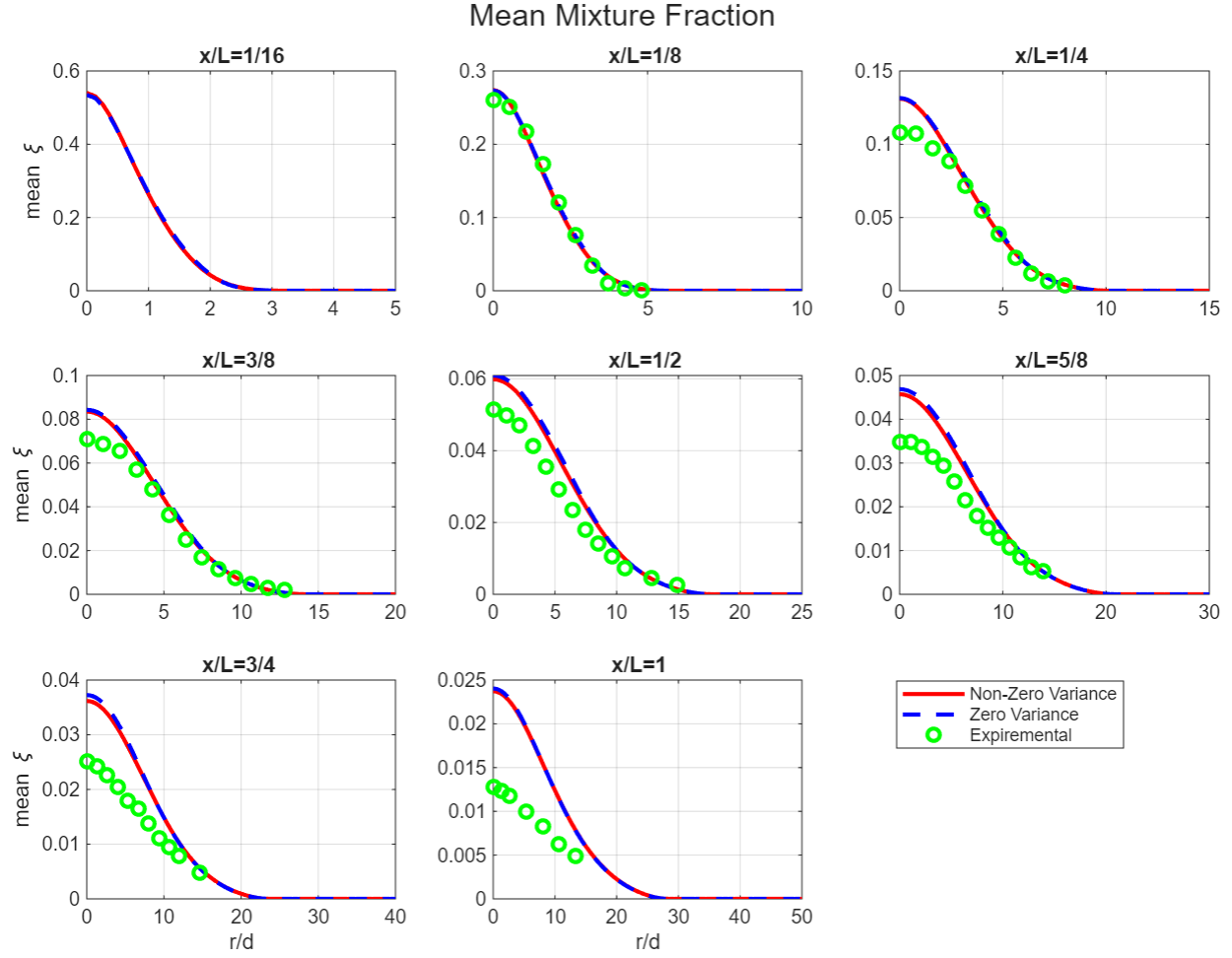


Figure 5 Mean Mixture Fraction versus r/d at Differing x/L

Fig. 5 shows the mean mixture fraction as a function of radial distance at differing horizontal distance locations. The non-zero variance model and the zero variance model follow each other well with the zero variance model having slightly larger values near the centerline and the horizontal distance is increased. The models consistently overestimate the mean mixture fraction especially near the centerline. The only location that matches well over the whole range of radial values is the $x/L=1/8$ case.

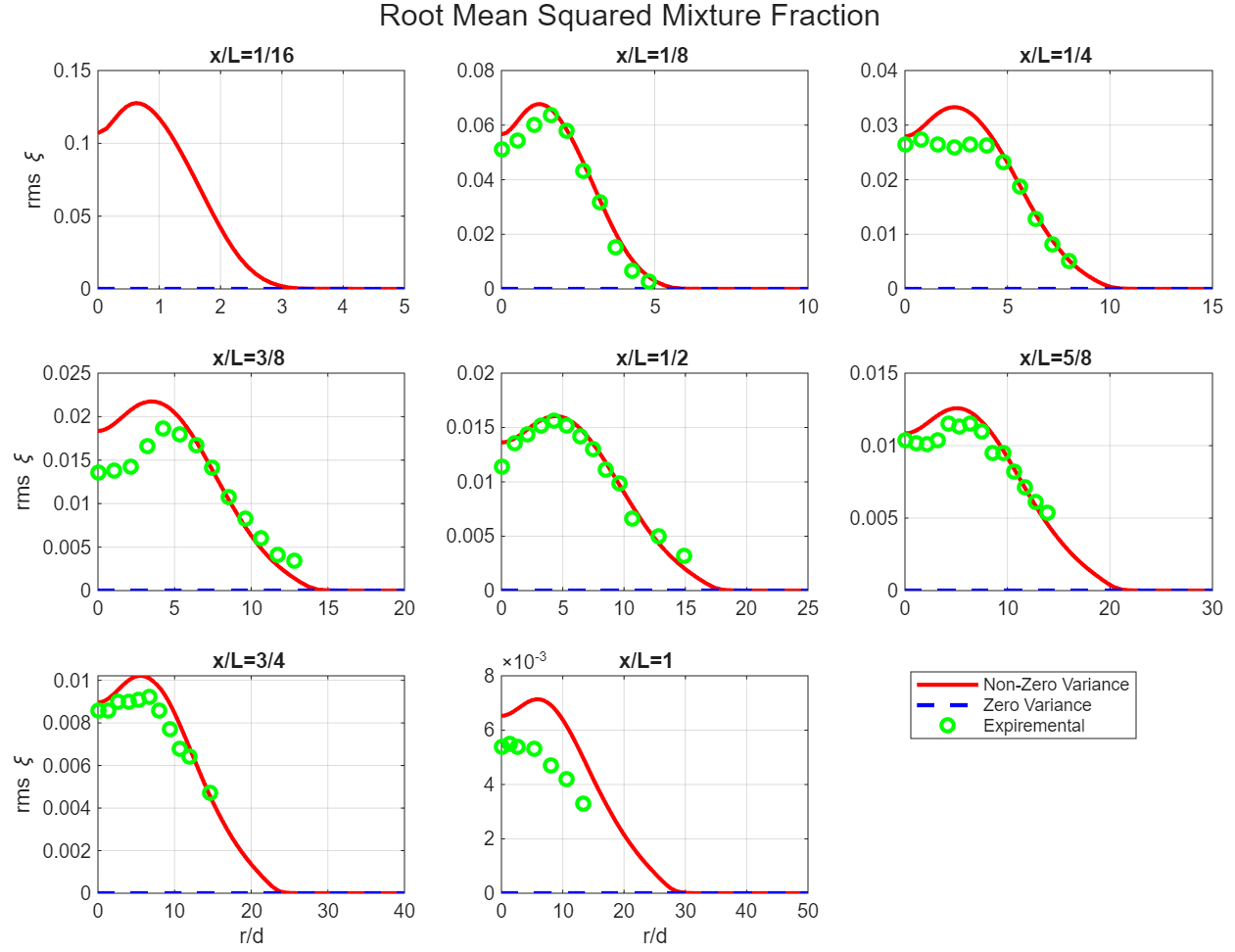


Fig. 6 shows the root mean squared mixture fraction as a function of radial distance at differing horizontal distance locations. The zero variance model has a value of 0 for each location which shows the Fluent model of zero variance is successful as the RMS mixture fraction strongly correlates with the variance as the RMS mixture fraction is defined as the square root of the variance. The non-zero variance model correlates well with the experimental results with the model overestimating the results at increasing horizontal distances especially near the centerline of the flame.

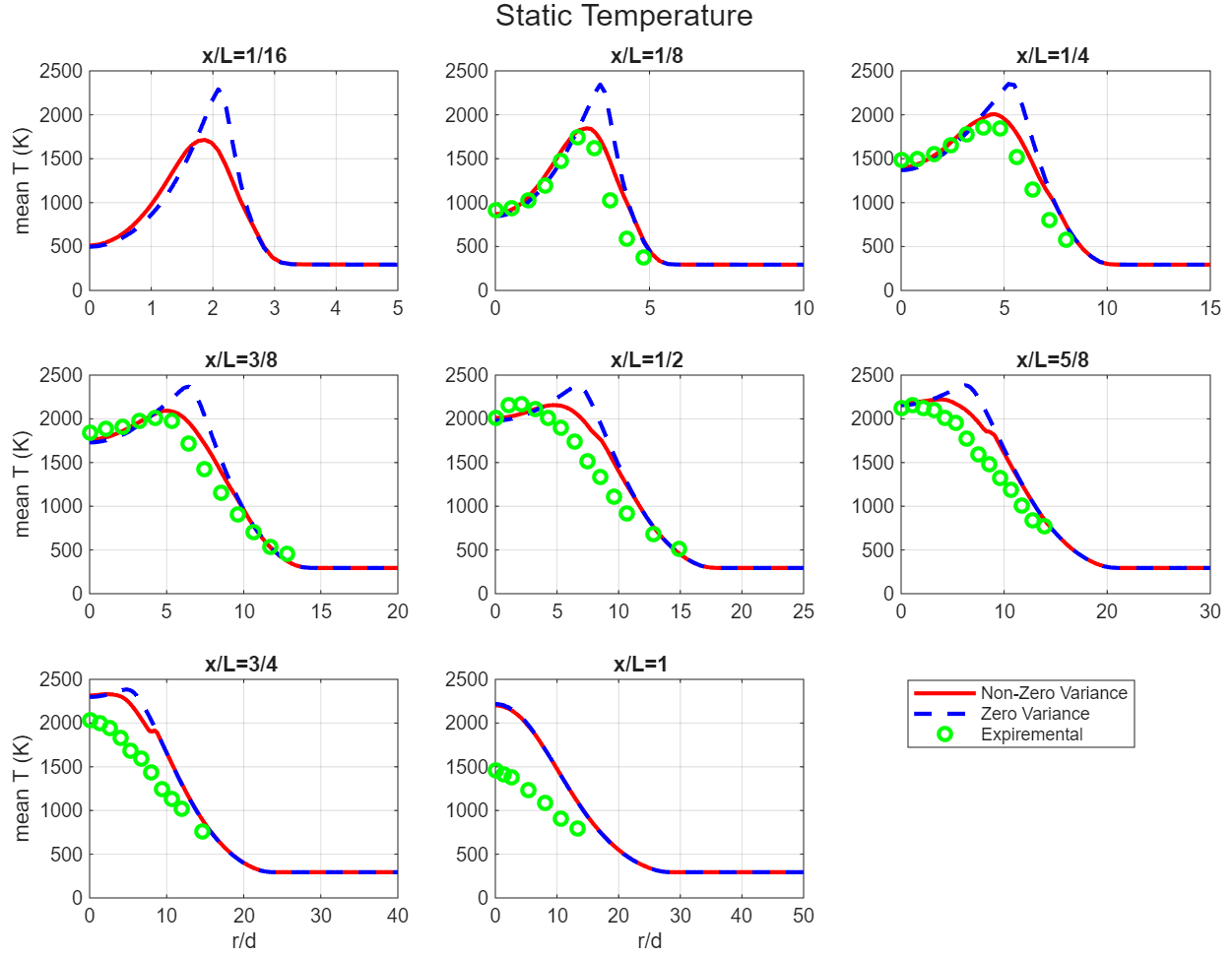


Figure 7 Static Temperature versus r/d at Differing x/L

Fig. 7 shows the static temperature as a function of radial distance at differing horizontal distance locations. The models differ near the flame inlet with the zero variance model having a much larger peak temperature than the non-zero variance model; however, they correlate better farther away from the flame front at increasing horizontal locations. Overall, the non-zero variance model is a better fit for the experimental data; however, both models overestimate the experimental results at increasing horizontal distances, especially near the centerline.

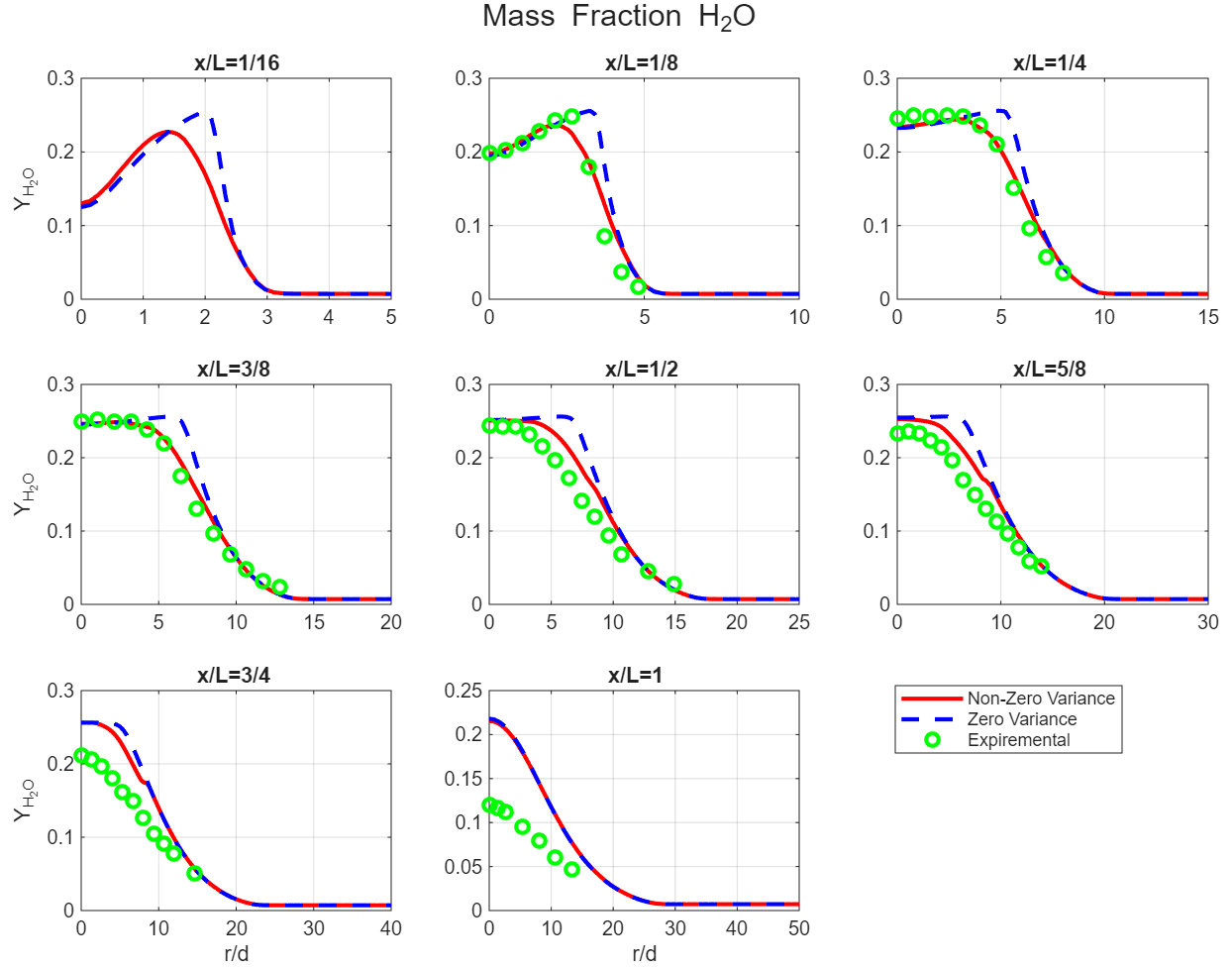


Fig. 8 shows the mass fraction of H_2O as a function of radial distance at differing horizontal distance locations. The zero variance model estimates a larger mass fraction at a larger radial distance compared to the non-zero variance model. However, these models tend to correlate better at increasing horizontal locations. The non-zero variance model tends to model the experimental results better than the zero variance model does; however, both models overestimate the experimental results especially near the centerline at increasing horizontal distances.

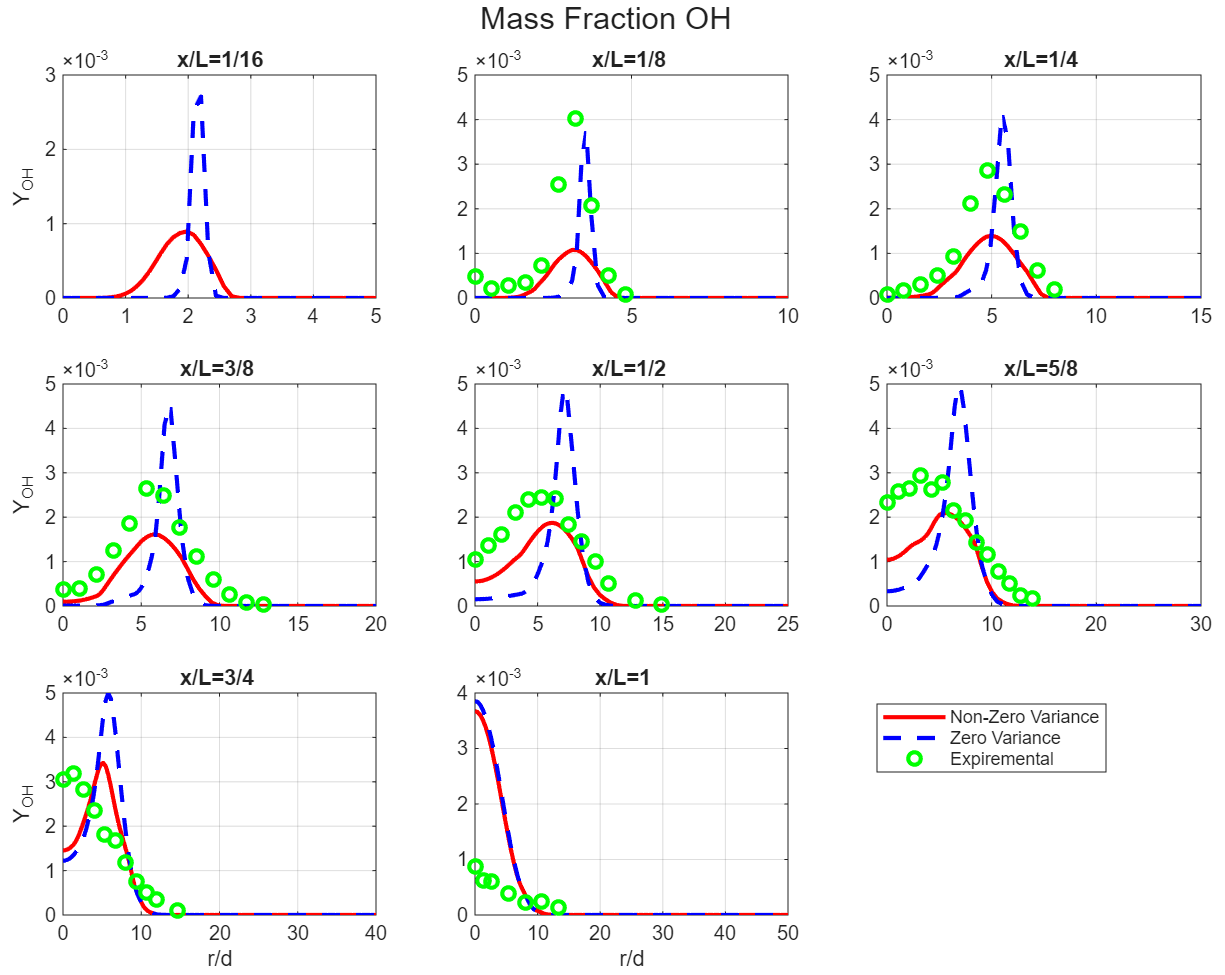


Fig. 9 shows the mass fraction of OH as a function of radial distance at differing horizontal distance locations. The zero variance model has a much greater peak mass fraction compared to the non-zero variance model; however, the models get closer at increasing horizontal distance locations. The experimental data tends to follow the zero variance model near the flame inlet, but at increasing horizontal distance, the zero variance model overestimated the data, and the data fits better with the non-zero variance model.

Table 1 Maximum Percent Error for all Outputs

	Axial Velocity	Root Mean Squared Velocity	Mean ξ	Root Mean Squared ξ	Static Temperature	Mass Fraction H ₂ O	Mass Fraction OH
Non-Zero Variance - Zero Var	6.9556	6.9556	3.7705	57.676	14.856	16.23	31938.0
Non-Zero Variance - Experimental	97.878	97.878	181.58	60.521	70.859	89.329	459.68
Zero Variance - Experimental	102.46	102.46	173.57	72.28	83.679	103.11	489.85

Table 1 shows the maximum percent errors between the two models and the experimental data for each of the outputted results. Both the models have lower percent errors with each other except for the mass fraction of OH which has extremely larger errors. This means that the models are most different for the mass fraction of OH. The percent errors show that the non-zero variance model is a better fit for the data for everything except for the mean mixture fraction. However, the maximum percent errors are still large for both the models, but that is to be expected because the models have a harder time near the flame front as there are large gradients in both mass fraction and temperature.

1.3 Sources of Discretization Error – QA(6)

There are multiple sources of discretization errors. Table 2 shows each discretization error, and how each one can be reduced.

Table 2 Sources of Errors

Sources of Error	How to Reduce
Spatial discretization	Refine mesh, higher-order schemes
Temporal discretization	Reduce time step, higher-order time integration
Source term approximation	Fine mesh, implicit solvers
Gradient calculation	Use higher-order gradient schemes, refine mesh
Interpolation	Higher-order face schemes, improve mesh quality
Boundary condition	Refine near-boundaries, use accurate boundary condition schemes
Numerical diffusion	Higher-order schemes, refine mesh, reduce skewness

1.4 Assumptions of Equilibrium Model – QA(7)

The equilibrium model uses many assumptions. Table 3 shows these assumptions and their meanings.

Table 3 Principle Assumptions in Equilibrium Model

Assumption	Meaning
Chemical Equilibrium	All chemical reactions reach thermodynamic equilibrium instantly, and species concentrations are only determined by local pressure and temperature
Neglect Finite-Rate Chemistry	Reaction rates are infinitely fast (does not account for reaction time scales)
Steady-State	Flow and combustion are steady, which allows for equilibrium species to be calculated at each point without time-dependent kinetics
Well-Mixed	The mixture is sufficiently mixed locally so that temperature and species are uniform at a point
Neglect of Transport-Limited Chemistry	Species diffusion, heat conduction, and turbulent mixing are much slower than the chemical reactions are (flame is chemically thin)
No Detailed Intermediate Species Tracking	Only major products are important, and the intermediate/residuals are inferred from equilibrium, not reaction paths

These assumptions are least likely to hold true near the flame front due to the steep temperature and species gradients. They are also least likely to hold near low-temperature areas and regions with strong turbulence as well as near stoichiometric boundaries.

2 Flamelet Model

2.1 Flamelet Information – QB(1)

Using user defined flamelet parameters with the number of grid points in flamelet set to 32 and the maximum number of flamelets set to 100, the number of flamelets generated is 28. The maximum scalar dissipation is 126 s^{-1} . This is found with the information that the initial scalar dissipation (χ_0) is 0.01 s^{-1} , the scalar dissipation multiplier (A) is 10, and the scalar dissipation step (n) is 5 s^{-1} . These values are default by Fluent. This means the scalar dissipation is akin to a geometric series. Eq.(1) shows the formula for the maximum scalar dissipation (χ_{max}).

$$\chi_{max} \approx \chi_0 A^{n-1} \quad (1)$$

If Eq.(1) is used, an approximate value of 100 s^{-1} is achieved for χ_{max} . The actual maximum value is 126 s^{-1} , which is found in fluent. However, this is close to the value of 100 that is

approximated. The maximum temperature in this flamelet is 2352.471 K and it occurs at scalar dissipation of 0.0100 and a mixture fraction of 0.03012.

2.2 Mixture Fraction and Scalar Dissipation – QB(2)

The scalar dissipation value will greatly affect both the temperature and mass fractions. The lowest value of scalar dissipation is 0.01 s^{-1} , and the highest value of scalar dissipation is 126 s^{-1} as shown in section 2.1 above.

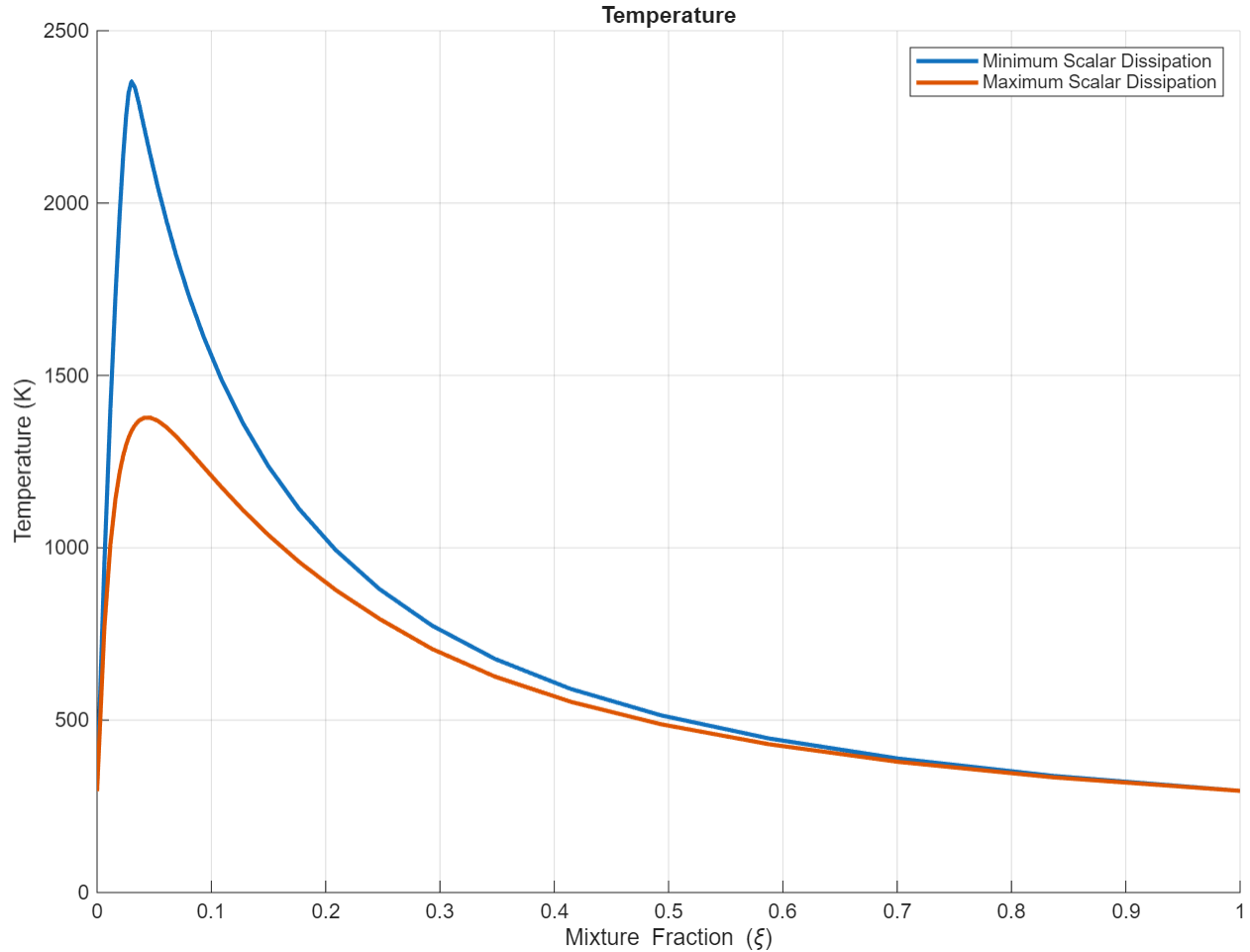


Figure 10 Temperature versus Mixture Fraction

Fig. 10 shows the temperature versus mixture fraction for both minimum and maximum scalar dissipation values. The minimum scalar dissipation value has a much larger peak temperature at roughly the same mixture fraction (0.05) as the maximum scalar dissipation does. The difference in temperature between the two models becomes less at increasing mixture fractions. Which means at larger mixture fractions, the value of scalar dissipation chosen becomes less important if only temperature is desired.

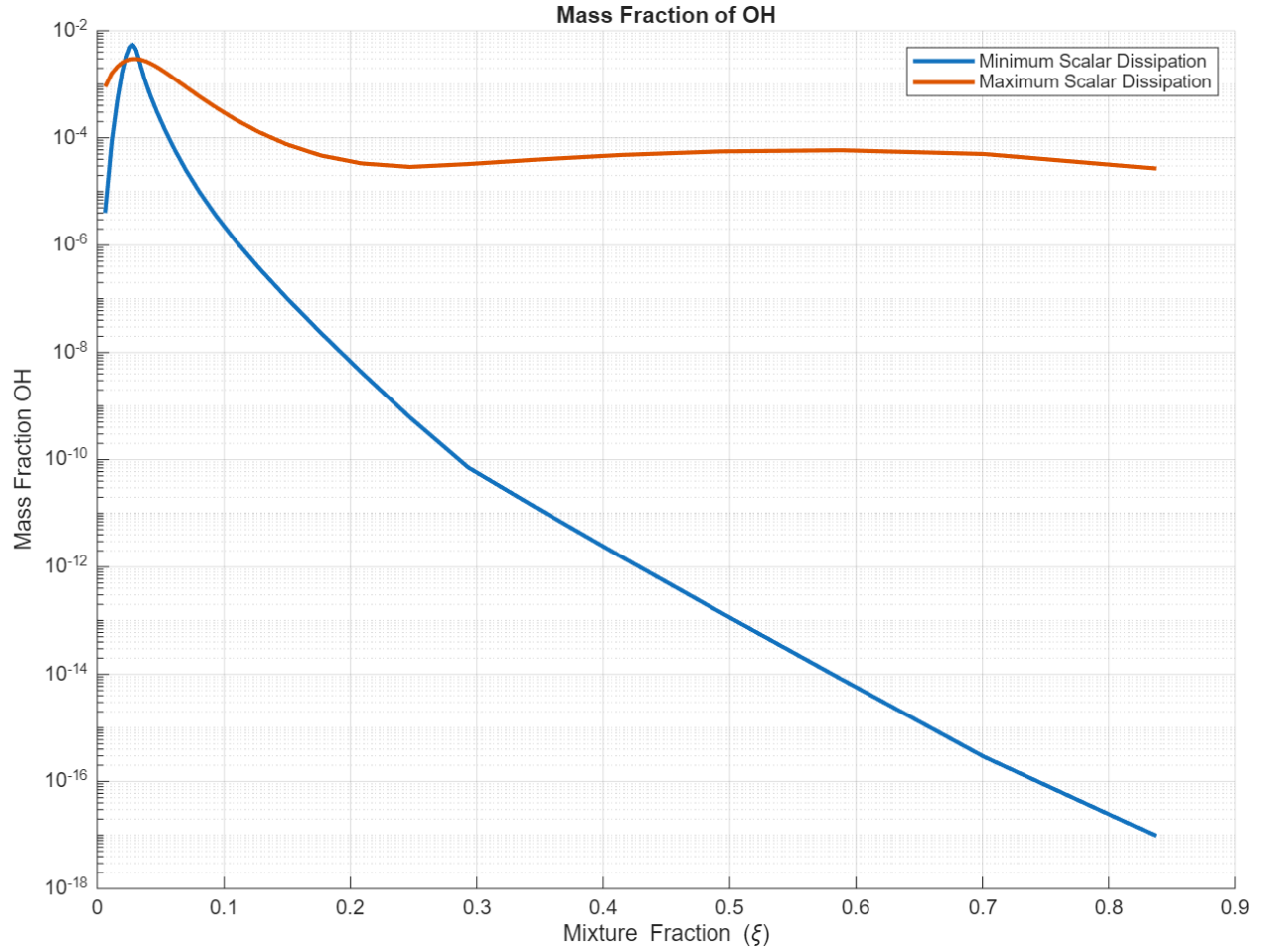


Figure 11 Mass Fraction of OH versus Mixture Fraction

Fig. 11 shows the mass fraction of OH versus the mixture fraction at the minimum scalar dissipation value and the maximum scalar dissipation value. The minimum scalar dissipation value has a larger peak mass fraction, but it occurs at the same mixture fraction as the maximum model. However, the mass fraction as increasing mixture fractions is much smaller for the minimum scalar dissipation model meaning the OH behaves more like a radical in this model as the mass fraction goes below 10^{-6} at a mass fraction of 0.11 while the maximum scalar dissipation model has a mass fraction of OH that stays above 10^{-5} throughout the mixture fraction range.

2.3 Equilibrium versus Flamelet Model – QB(3)

The equilibrium and flamelet models are slightly different in the terms of how the species are calculated, but the flamelet model tends to be more accurate in terms of a non-premixed flame which is what is analyzed here.

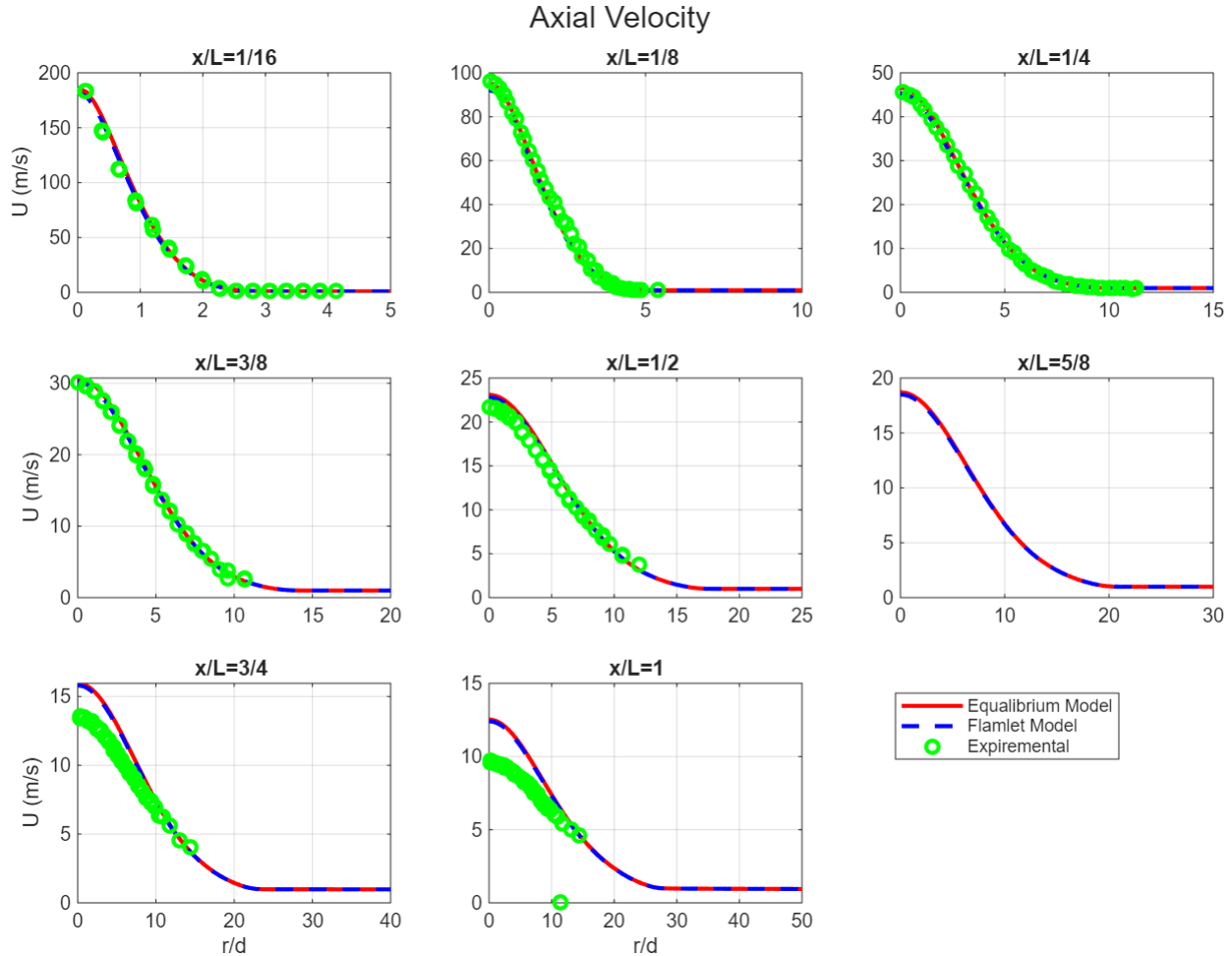


Figure 12 Axial Velocity versus r/d at Differing x/L

Fig. 12 shows the axial velocity as a function of radial distance at differing horizontal distance locations. The equilibrium model and the flamelet model both follow very similar paths with the flamelet model having slightly smaller axial velocity as the horizontal distance increases. The experimental results follow both models well; however, the correlation gets less at an increasing horizontal distance as the models overestimate the axial velocity, especially near the centerline of the flame.

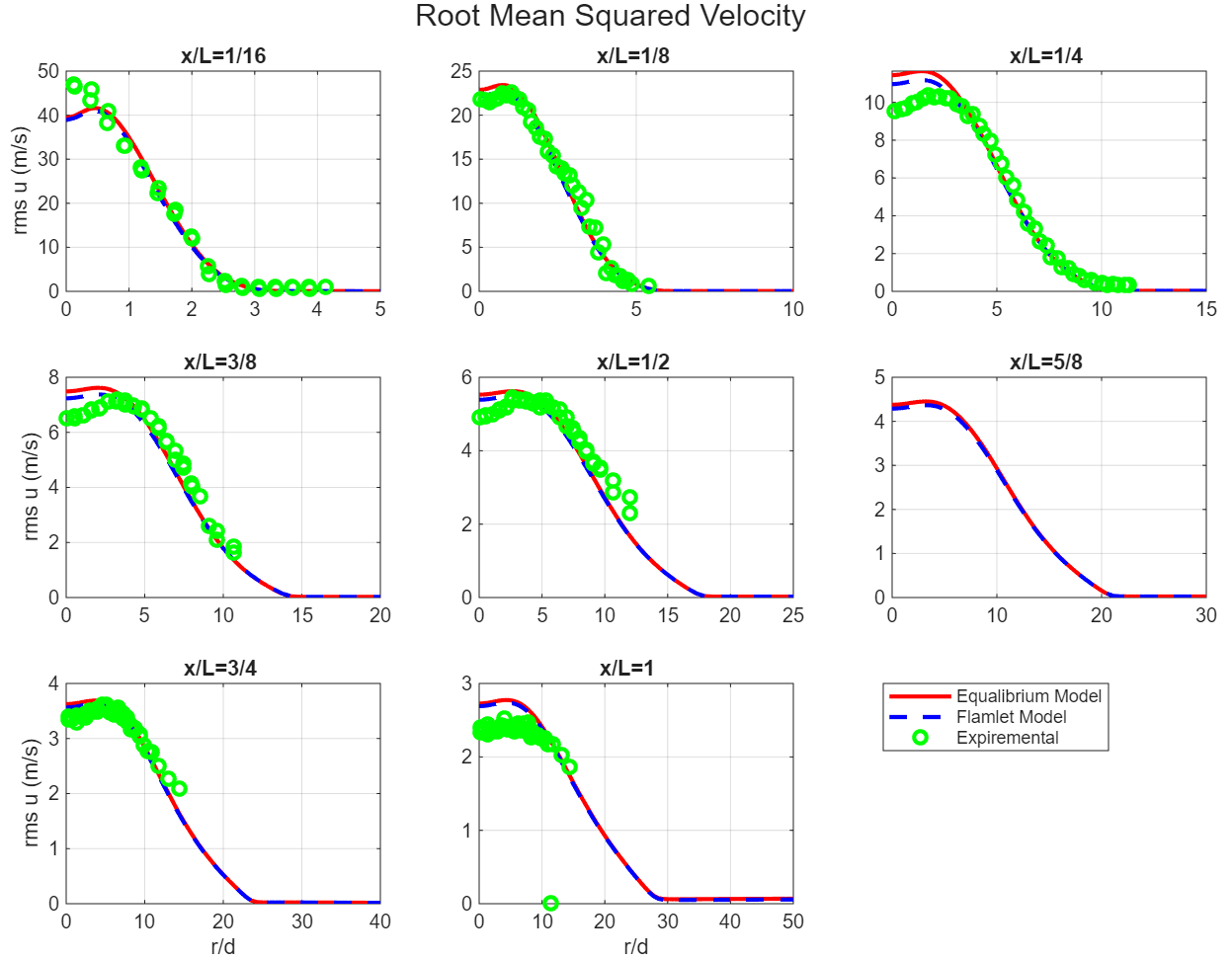


Fig. 13 shows the root mean squared velocity as a function of radial distance at differing horizontal distance locations. The equilibrium model and the flamelet model both follow very similar paths with the flamelet model having slightly smaller RMS velocity as the horizontal distance increases especially near the centerline of the flame with the maximum distance being around the middle of the flame. The experimental results are consistently under the model results which means the models are overestimating the results, especially near the centerline. The exception to this is near the flame inlet where at the centerline (the flame inlet), the models underestimate the RMS velocity.

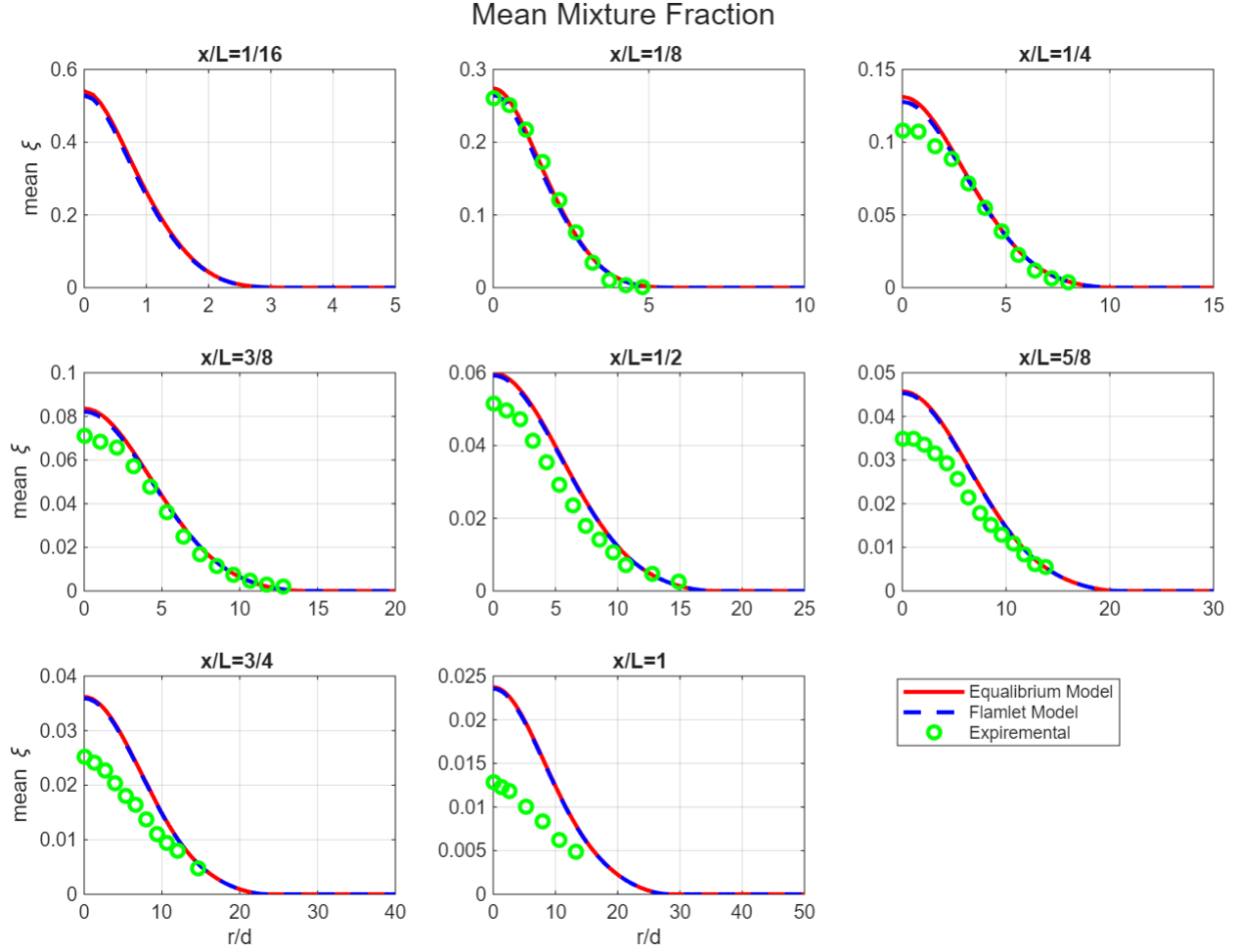


Figure 14 Mean Mixture Fraction versus r/d at Differing x/L

Fig. 14 shows the mean mixture fraction as a function of radial distance at differing horizontal distance locations. The equilibrium model and the flamelet model follow each other well with the flamelet model having slightly smaller values near the centerline and the horizontal distance is decreased. This means the models are most different near the flame front at the centerline, but even then, they are very close. The models consistently overestimate the mean mixture fraction especially near the centerline. The only location that matches well over the whole range of radial values is the $x/L=1/8$ case.

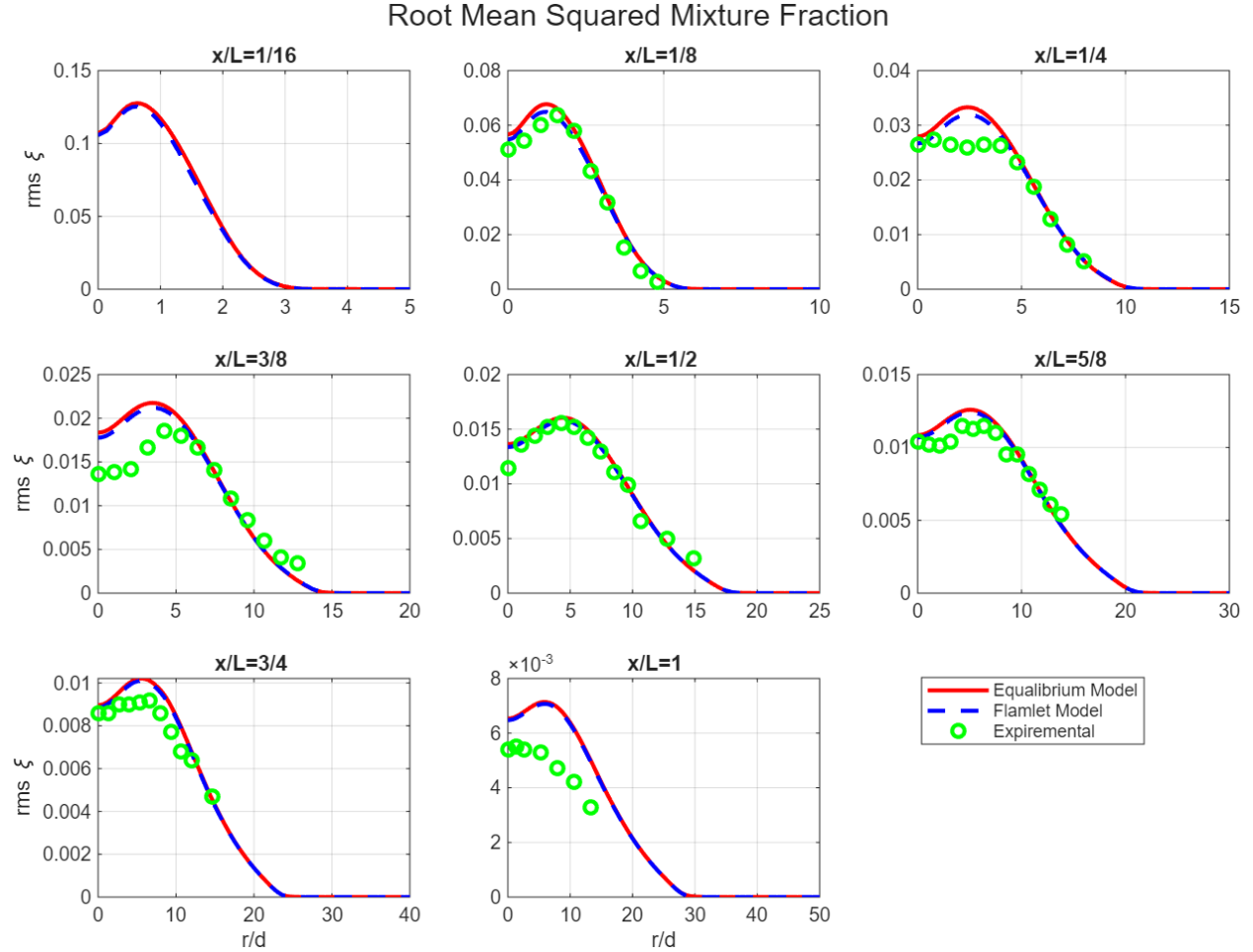


Fig. 15 shows the root mean squared mixture fraction as a function of radial distance at differing horizontal distance locations. The equilibrium model and the flamelet model are very similar; however, they are most different near the flame inlet (at decreasing horizontal locations) especially near where the maximum RMS mixture fraction occurs with the flamelet model having a lower value. The models tend to overestimate the experimental data; however, the flamelet model tends to do a better job at modeling it at certain horizontal locations.

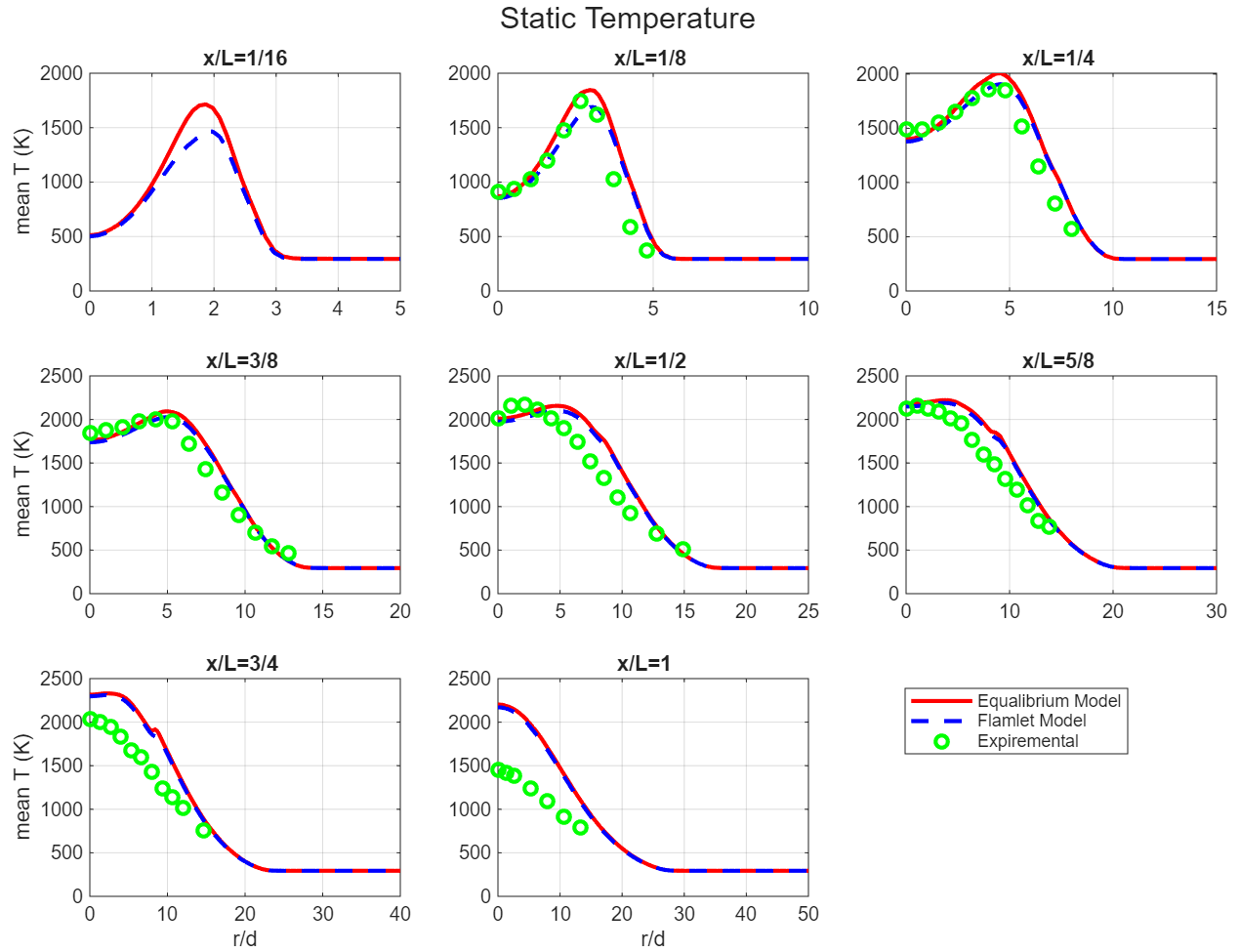


Figure 16 Static Temperature versus r/d at Differing x/L

Fig. 16 shows the static temperature as a function of radial distance at differing horizontal distance locations. The models differ near the flame inlet with the equilibrium model having a larger peak temperature than the flamelet model; however, they correlate better farther away from the flame front at increasing horizontal locations. Overall, the flamelet model is a better fit for the experimental data; however, both models overestimate the experimental results at increasing horizontal distances, especially near the centerline.

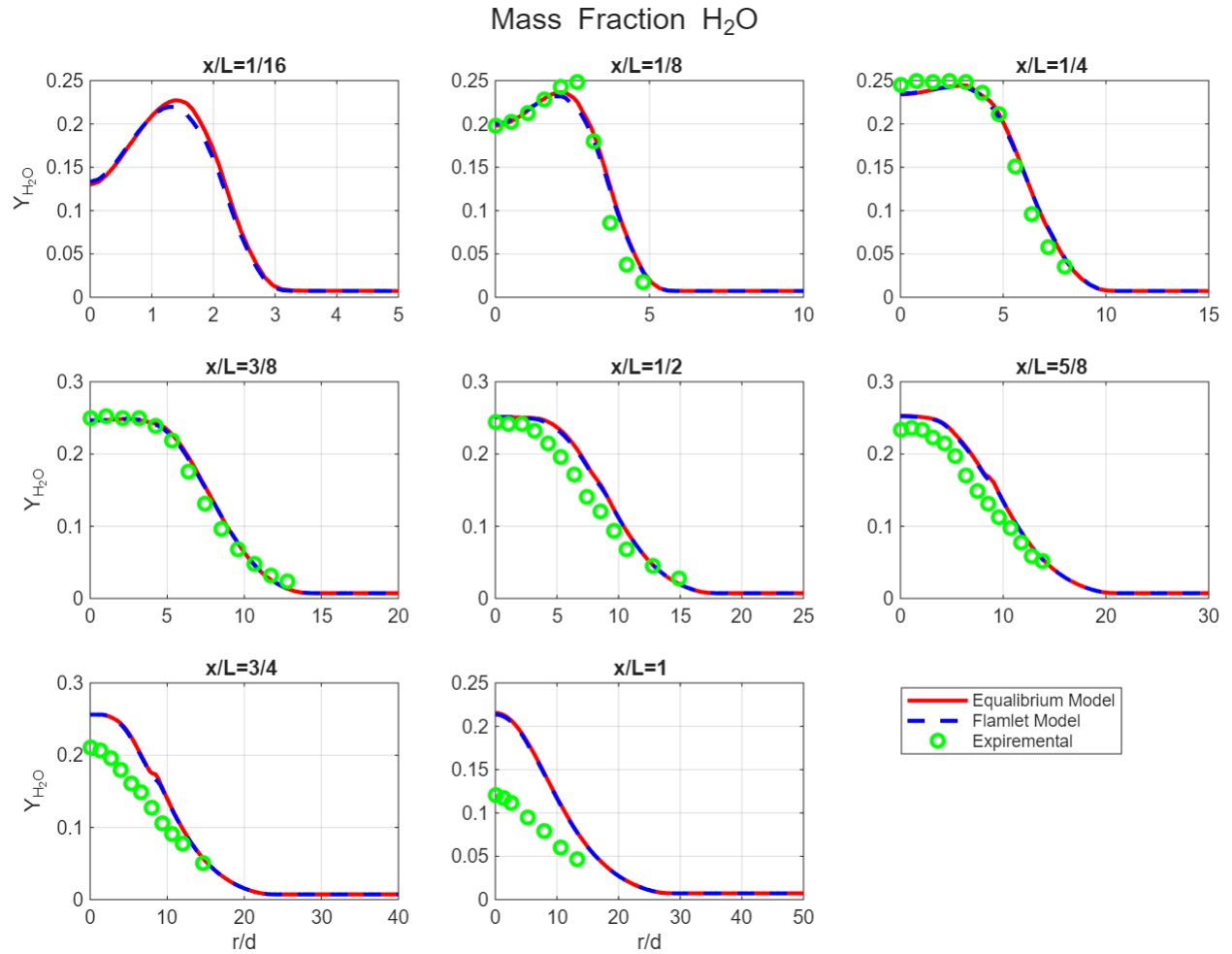


Figure 17 Mass Fraction of H_2O versus r/d at Differing x/L

Fig. 17 shows the mass fraction of H_2O as a function of radial distance at differing horizontal distance locations. The equilibrium model and the flamelet model both have a very strong correlation which the flamelet model estimates a slightly smaller peak mass fraction at the inlet of the flame. The models estimate the experimental results well near the inlet; however, as the horizontal distance increases, the models tend to overestimate the experimental data especially near the centerline.

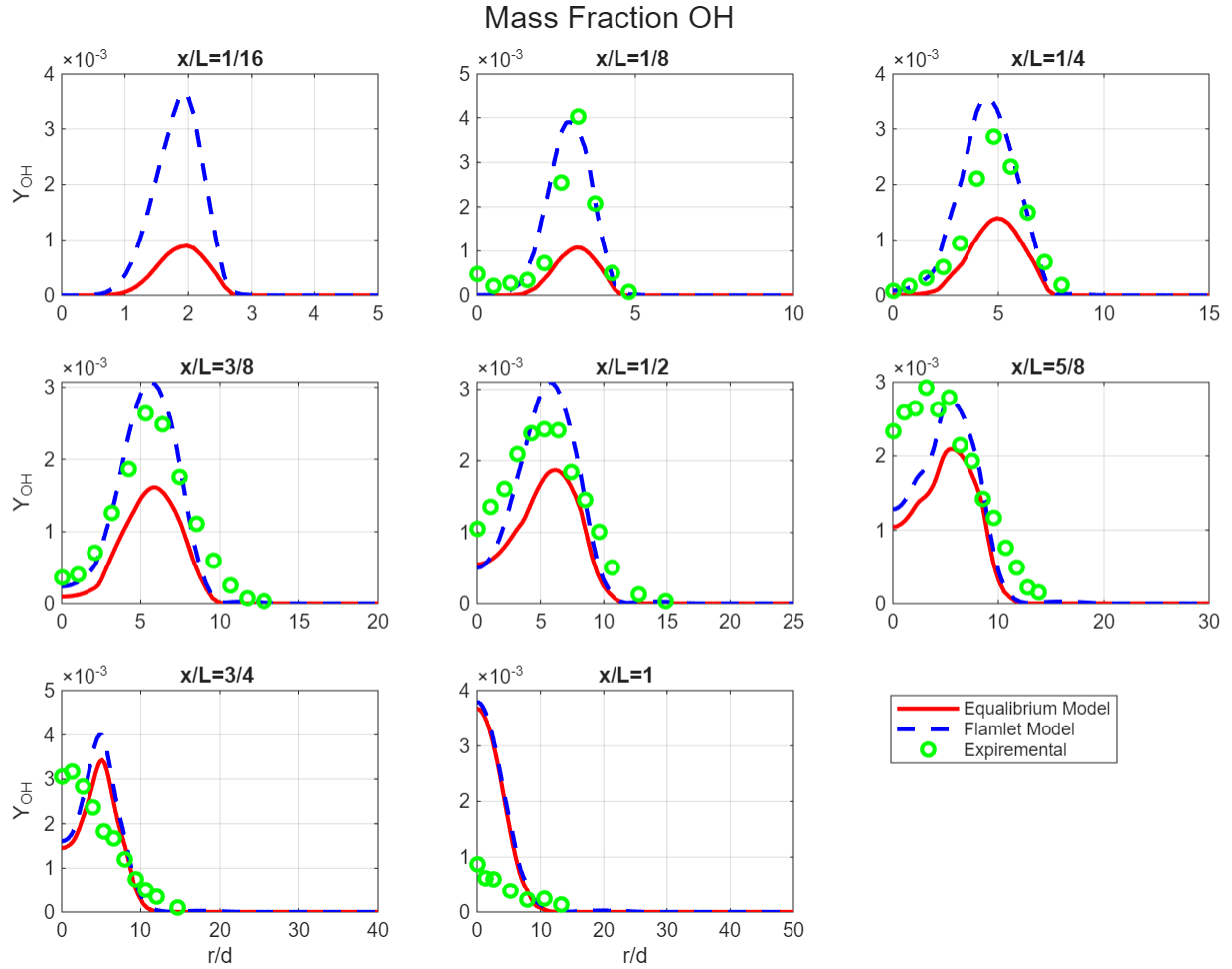


Fig. 18 shows the mass fraction of OH as a function of radial distance at differing horizontal distance locations. The flamelet model has a much greater peak mass fraction compared to the equilibrium model; however, the models get closer at increasing horizontal distance locations. The experimental data tends to follow the flamelet model near the flame inlet, but at increasing horizontal distance, flamelet model overestimates the data, and the data fits better with the equilibrium model. At the end of the flame-tube, the models both greatly overestimate the experimental results.

Table 4 Maximum Percent Error for all Outputs

	Axial Velocity	Root Mean Squared Velocity	Mean ξ	Root Mean Squared ξ	Static Temperature	Mass Fraction H ₂ O	Mass Fraction OH
Equilibrium-Flamelet	17.4778	17.4778	4.7677	5.4399	11.6529	3.9217	100.0000
Equilibrium-Experimental	97.8777	97.8777	181.5774	60.5212	70.8592	89.3293	459.6814
Flamelet - Experimental	94.1257	94.1257	168.7637	58.3285	65.9047	87.3294	477.5395

Table 4 shows the maximum percent errors between both the models and the experimental results for all the outputs. Between the equilibrium and flamelet models, the percent errors are low except for the mass fraction of OH. This means that the models are very similar to each other for most of the output values. The flamelet model has lower percent errors with the experimental results than the equilibrium model does which means it is a better model compared to the equilibrium model. However, the percent errors are very close.

2.4 Mixture Fraction and Mean Scalar Dissipation Model – QB(4)

The model equation for the mixture fraction variance (g) is shown in Eq.(2).

$$g \equiv \tilde{\xi}^{1/2} \quad (2)$$

Eq.(3) shows the model for the mean scalar dissipation ($\tilde{\chi}$)

$$\tilde{\chi} = C_d g \frac{\varepsilon}{k} \quad (3)$$

Fig. 19 shows the mean scalar dissipation at differing horizontal distances as a function of radial distance from the centerline of the flame.

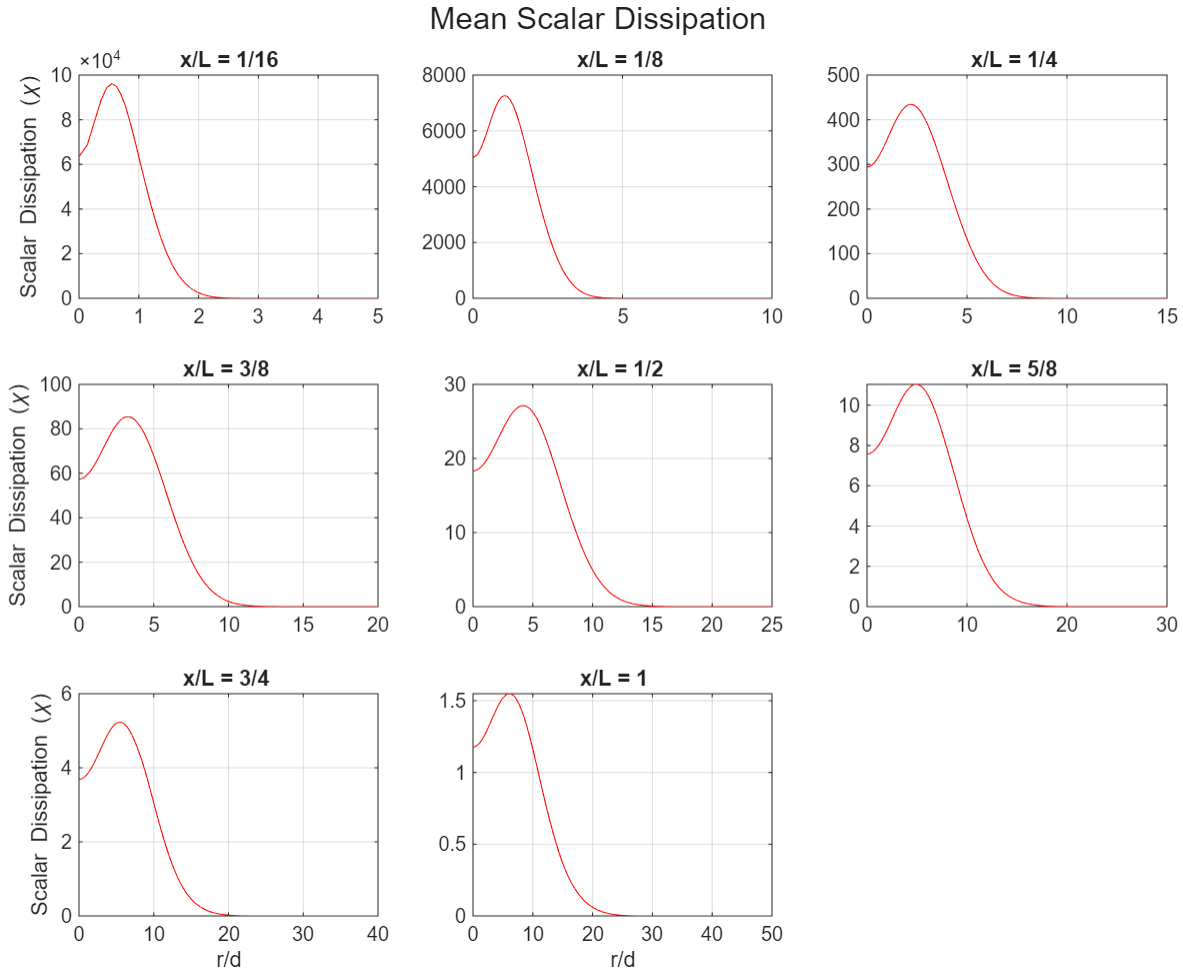


Figure 19 Mean Scalar Dissipation versus r/d at Differing x/L

The maximum mean scalar dissipation ($\tilde{\chi}$) is 96140.8, and it occurs at a r/d of 0.546992 (roughly halfway from the centerline of the flame and the edge of the flame tube). It occurs at a $x/L = 1/16$ which is very close to the flame inlet. The centerline ($x/L=1/2$) does vary with axial distance as approximately a power law. This is shown in Fig. 19 especially from an axial distance of 5 to 20 due to the shape of the graph.